# HONG KONG BAPTIST UNIVERSITY
# COURSE OUTLINE

## 1. COURSE TITLE

Web and Mobile Programming

## 2. COURSE CODE

COMP7270

## 3. NO. OF UNITS

3 Units

## 4. OFFERING DEPARTMENT

Master of Science in Data Analytics and Artificial Intelligence

## 5. PREREQUISITES

Postgraduate Student Standing

## 6. MEDIUM OF INSTRUCTION

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## 7. AIMS & OBJECTIVES

To equip students with information and communication technology skills to design and implement web and mobile applications for data acquisition, storage and visualization.

## 8. COURSE CONTENT

I. Getting Started with Webpage Development
- HTML and CSS
- Responsive web design
- Document object model (DOM) and client-side JavaScript

II. Web Application Development
- Hypertext Transfer Protocol (HTTP)
- Server-side programming
- Document oriented database
- AJAX techniques and Restful web services

III. Mobile Application Development

- Front-end JavaScript framework, such as Vue.js
- Hybrid mobile application development
- Software architecture patterns

IV. Selected Topics in System Development
- Version control system for software development
- Basic data visualization for web and mobile platforms
- Usage of pre-trained AI models on web and mobile applications

## 9. COURSE INTENDED LEARNING OUTCOMES (CILOs)

| CILO | By the end of the course, students should be able to: |
|---|---|
| CILO 1 | Explain the knowledge of web standards including markup languages, stylesheets and scripts |
| CILO 2 | Explain the knowledge of web protocols and options in data transaction over networks |
| CILO 3 | Explain the knowledge of software architectures and components of web and mobile applications |
| CILO 4 | Design and implement a web application for presenting and managing a data-driven system |
| CILO 5 | Design and implement a mobile application for presenting and managing a data-driven system |

## 10. TEACHING & LEARNING ACTIVITIES (TLAs)

| CILO alignment | Type of TLA |
|---|---|
| 1-3 | Lectures are conducted to teach concepts, programming techniques, and database techniques for web and mobile application development. They provide theoretical knowledge and foundational understanding of the subject matter. |
| 4-5 | Laboratory sessions are hands-on practical sessions conducted in computer labs. Students can apply their knowledge and skills acquired from lectures and tutorials to work on exercises, experiments, and projects related to web and mobile application development. |

## 11. ASSESSMENT METHODS (AMs)

| Type of Assessment Methods | Weighting | CILOs to be addressed | Description of Assessment Tasks |
|---|---|---|---|
| Assignments and quizzes | 40 % | 1-5 | The assignments and in-class quizzes aim to assess students' understanding of concepts, |

| | | | |
|---|---|---|---|
| | | | programming techniques, and database skills required for developing web and mobile applications. The assignments will provide students with practical tasks to demonstrate their knowledge and application of these concepts. |
| Group Project | 20 % | 1-5 | The group project serves as an evaluation of students' acquisition of knowledge and software skills related to web and mobile application development. This project will require students to collaborate in teams to design and implement a web or mobile application, showcasing their ability to apply the learned concepts in a real-world context. |
| Examination | 40 % | 1-5 | The final examination is designed to assess the extent to which students have achieved their intended learning outcomes. The examination will primarily consist of analysis and skills-based questions, evaluating students' ability to apply their knowledge to web and mobile application development. The exam will test their understanding of the course material and their proficiency in practical application. |

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